UNIT - I : STRUCTURE OF METALS AND CONSTITUTION OF ALLOYS

1. Introduction

* Metals are crystalline solids: atoms are arranged in a regular, repeating 3-D pattern called the SPACE LATTICE.
* Unit cell = smallest group of atoms which, repeated in 3-D, builds the whole crystal. Lattice parameters: a, b, c (edges) and angles alpha, beta, gamma.
* Amorphous solids (glass) have no long-range order. Most metals crystallise into BCC, FCC or HCP.
* Important terms: Coordination Number (CN) = number of nearest neighbour atoms; Atomic Packing Factor (APF) = volume of atoms in cell / volume of cell.

$$APF = (n \times 4/3\ \pi r^3) / V_{cell}$$

2. Crystallization of Metals

When a molten metal cools below its freezing point, solidification (crystallization) takes place in 2 steps:

* (i) NUCLEATION - formation of tiny stable solid nuclei in the liquid.
* (ii) GROWTH - atoms from the liquid attach to the nuclei, crystals grow until they meet each other.

Nucleation

* Homogeneous nucleation: nuclei form by themselves in pure liquid; needs large undercooling (delta T).
* Heterogeneous nucleation: nuclei form on impurities / mould walls; needs little undercooling. Used in practice (inoculants / grain refiners such as Ti, B in Al).
* Free energy change for a spherical nucleus of radius r :

$$\Delta G = -(4/3)\ \pi r^3 \Delta G_v + 4\ \pi r^2\ \gamma$$

* Volume term (negative) favours growth, surface-energy term (positive) opposes it. Maximum of Delta G occurs at the CRITICAL RADIUS r^*.

$$r^* = 2\ \gamma\ T_m / (\Delta H_f \cdot \Delta T)$$

* Embryo with $r < r^*$ dissolves; nucleus with $r > r^*$ is stable and grows. Greater

undercooling -> smaller r^* -> more nuclei -> FINE grains.

Cooling curve of a pure metal

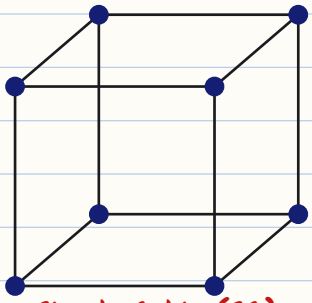
* Temperature falls, undercooling occurs slightly below freezing point, then rises back (latent heat released) and stays constant (thermal arrest) till solidification is complete; then falls again.

Growth - Dendritic growth

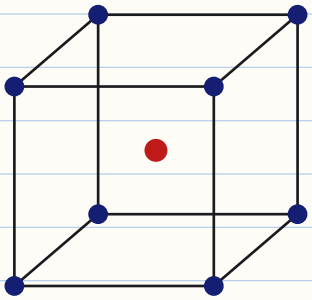
* Crystals grow fastest along certain directions, forming tree-like DENDRITES (primary arm with secondary and tertiary arms). Liquid between the arms finally freezes; the dendrites meet and form GRAINS separated by GRAIN BOUNDARIES.

Fast cooling -> more nuclei -> fine grains (stronger, tougher). Slow cooling -> few nuclei -> coarse grains (weaker).

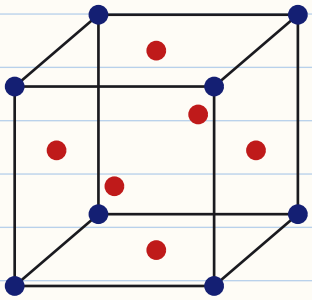
3. Crystal Structures and Packing Factor



Simple Cubic (SC)
n = 1, CN = 6



BCC
n = 2, CN = 8



FCC
n = 4, CN = 12

Property	SC	BCC	FCC	HCP
Atoms / cell (n)	1	2	4	6
Coord. number	6	8	12	12
a in terms of r	a = 2r	a = 4r/sqrt3	a = 4r/sqrt2	a = 2r
APF	0.52	0.68	0.74	0.74
Examples	Po	Fe(a),Cr,W,Mo	Al,Cu,Ni,Ag,Au	Mg,Zn,Ti,Cd

(a) Simple Cubic (SC)

* Atoms touch along the cube edge -> a = 2r. Atoms per cell = 8 x 1/8 = 1.

$$APF = (1 \times 4/3\ \pi r^3) / (2r)^3 = \pi/6 = 0.52$$

(b) Body Centred Cubic (BCC)

* Atoms touch along the body diagonal: sqrt3 a = 4r -> a = 4r/sqrt3. Atoms per cell = 8 x 1/8 + 1 = 2. CN = 8.

$$APF = (2 \times 4/3\ \pi\ r^3) / (4r/\sqrt{3})^3 = \sqrt{3}\ \pi / 8 = 0.68$$

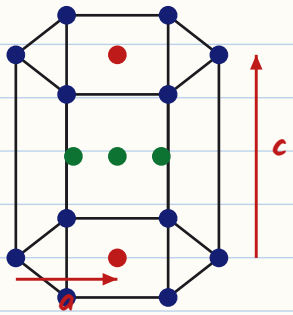
(c) Face Centred Cubic (FCC)

* Atoms touch along the face diagonal: sqrt2 a = 4r -> a = 4r/sqrt2 = 2 sqrt2 r. Atoms per cell = 8 x 1/8 + 6 x 1/2 = 4. CN = 12.

$$APF = (4 \times 4/3\ \pi\ r^3) / (2\sqrt{2}\ r)^3 = \pi / (3\sqrt{2}) = 0.74$$

* FCC has ABCABC... stacking of close packed planes {111}.

(d) Hexagonal Close Packed (HCP)



HCP unit cell

Atoms: 12 corners x 1/6 = 2

2 face-centre atoms x 1/2 = 1

3 inside atoms x 1 = 3 Total n = 6

CN = 12 , c/a = 1.633 (ideal)

* Stacking ABABAB... of close packed basal planes. a = 2r, ideal c/a = sqrt(8/3) = 1.633.
* Volume of HCP cell = (3 sqrt3 / 2) a^2 c.

$$APF = (6 \times 4/3\ \pi\ r^3) / ((3\sqrt{3}/2)\ a^2\ c) = \pi / (3\sqrt{2}) = 0.74$$

FCC and HCP are both CLOSE PACKED structures (APF = 0.74, CN = 12); only the stacking sequence differs.

4. Linear Density and Planar Density

$$LD = \frac{\text{no. of atoms centred on direction vector}}{\text{length of the vector}}$$

$$PD = \frac{\text{no. of atoms centred on the plane}}{\text{area of the plane}}$$

Linear density examples

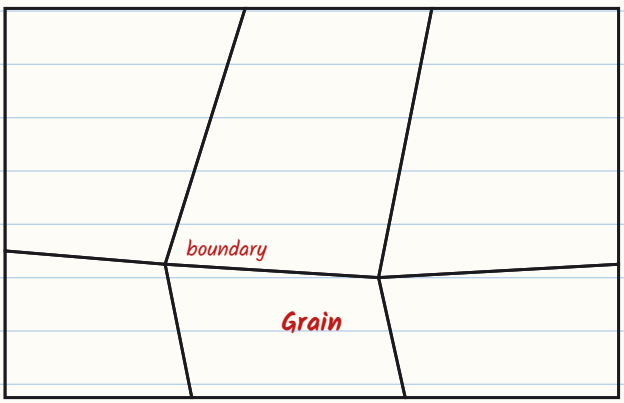
* SC [100]: 1 atom per edge length a -> LD = 1/a = 1/2r.
* BCC [111] (body diagonal): 2 atoms over length sqrt3 a = 4r -> LD = 2/(4r) = 1/2r (closest packed direction in BCC).
* FCC [110] (face diagonal): 2 atoms over length sqrt2 a = 4r -> LD = 2/4r = 1/2r (closest packed direction in FCC).
* FCC [100]: LD = 1/a = 1/(2 sqrt2 r) = 0.354/r.

Planar density examples

* SC (100) plane: 4 x 1/4 = 1 atom in area a^2 -> PD = $1/a^2$ = $1/4r^2$.
* BCC (110) plane: area = a x sqrt2 a; atoms = 4 x 1/4 + 1 = 2 -> PD = 2/(sqrt2 a^2) = 0.265/r^2 (most densely packed plane in BCC).
* FCC (111) plane: triangle of side sqrt2 a, atoms = 3 x 1/6 + 3 x 1/2 = 2 -> PD = 2 / ((sqrt3/2) a^2) = 4/(sqrt3 a^2) = 0.29/r^2 (closest packed plane of FCC, 6 neighbours in the plane).
* FCC (100): atoms = 4 x 1/4 + 1 = 2, area a^2 -> PD = $2/a^2$ = 0.25/r^2.

Slip occurs on planes of HIGHEST planar density along directions of HIGHEST linear density -> FCC metals (Cu, Al) are very ductile.

5. Grains, Grain Boundaries and Grain Size



Grain size (ASTM): N = 2^(n-1)

N = grains / sq. inch at 100x

n = ASTM grain size number

Larger n -> finer grains

Hall-Petch: sigma_y = sigma_0 + k d^(-1/2)

* GRAIN = single crystal region with uniform orientation. A metal piece (polycrystalline) is an aggregate of many grains.
* GRAIN BOUNDARY = narrow region (about 2-3 atom diameters) where neighbouring grains of different orientation meet; atoms are irregularly arranged, packing is poor.

Types

* Low angle (tilt) boundary: misorientation < 10 deg, made of array of edge dislocations. High angle boundary: misorientation > 15 deg.
* Twin boundary: mirror image across the boundary plane. Stacking fault: error in stacking sequence.

Effects of grain boundaries

* They block dislocation movement -> increase strength and hardness (Hall-Petch relation). Fine grain = high strength + toughness.
* They are high energy regions -> preferential etching (grain boundaries appear dark under the microscope), preferential corrosion, diffusion and precipitation.
* At high temperature (above 0.5 T_m) boundaries slide -> creep. Therefore coarse grained or single crystal blades are used in turbines.
* Boundaries scatter electrons -> slightly raise electrical resistivity.

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

Determination of grain size

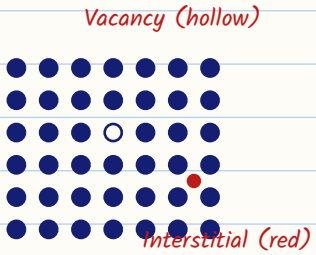
* ASTM method: sample polished, etched, examined at 100X magnification. Count grains per square inch N. Grain size number n :

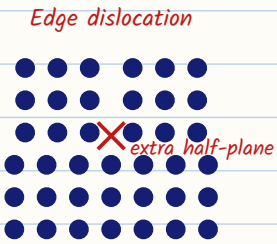
$$N = 2^{n-1}$$

* Comparison method: compare the micrograph with standard ASTM charts (n = 1 coarse to n = 8-10 fine).
* Intercept (Heyn) method: draw lines of known length L on the micrograph, count number of grain boundaries intersected P; mean intercept length = L/(P x M), M = magnification.
* Planimetric (Jeffries) method: draw a circle of known area, count grains inside (n_i) + half of grains cut by the circle ($n_c/2$); grains per mm^2 = f ($n_i + n_c/2$), f = Jeffries multiplier.

6. Imperfections in Crystals

Real crystals are never perfect. Imperfections (defects) control strength, ductility, conductivity and diffusion.





A. Point defects (0-dimensional)

* Vacancy: missing atom. Equilibrium concentration $n_v = N \exp(-Q_v/kT)$ (increases with temperature).
* Interstitial: extra atom squeezed into the space between lattice atoms (self interstitial or small impurity like C, N, H in iron).
* Substitutional impurity: foreign atom replaces a host atom.
* Frenkel defect: ion leaves its site and goes to interstitial position (vacancy + interstitial pair). Schottky defect: pair of cation + anion vacancies (ionic solids).

B. Line defects - Dislocations (1-dimensional)

* Edge dislocation: extra half plane of atoms inserted; dislocation line is the edge of this half plane. Burgers vector b is PERPENDICULAR to the dislocation line.
* Screw dislocation: crystal sheared so that atomic planes form a helical ramp. Burgers vector is PARALLEL to the line.
* Mixed dislocation: part edge, part screw.
* Burgers vector = magnitude and direction of lattice distortion (closure failure of Burgers circuit).
* Dislocation density = total length of dislocation line per unit volume (10^6 cm^{-2} annealed to 10^{12} cm^{-2} cold worked).

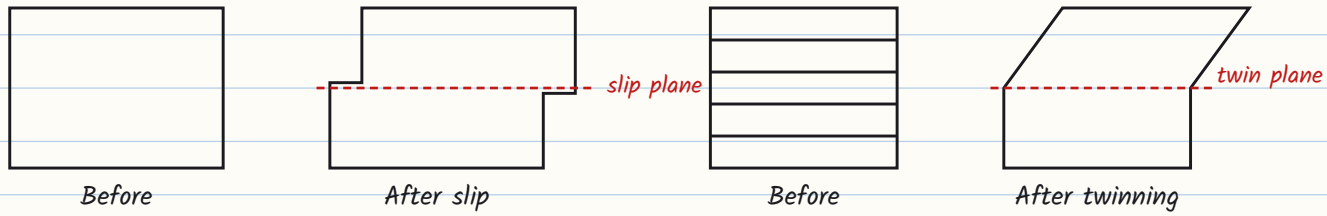
C. Surface (planar) defects (2-dimensional)

* External surface, grain boundaries, twin boundaries, stacking faults, phase boundaries.

D. Volume defects (3-dimensional)

* Pores, cracks, inclusions, voids, precipitates.

7. Slip and Twinning (Plastic deformation)



Slip

* Plastic deformation by sliding of one part of crystal over another along definite SLIP PLANES in definite SLIP DIRECTIONS. Slip plane + direction = SLIP SYSTEM.
* Slip occurs by movement of dislocations (needs much lower stress than shearing a perfect crystal - theoretical strength is about 1000 times larger than actual).
* Slip systems: FCC -> {111}<110> = 12 systems (very ductile). BCC -> {110}<111> (+{112},{123}) = 12 or more (fairly ductile). HCP -> basal (0001)<11-20> = only 3 at room temperature (brittle, e.g. Zn, Mg).
* Critical Resolved Shear Stress (Schmid's law): $\tau_R = \sigma \cos(\phi) \cos(\lambda)$.
* Slip lines are seen as fine steps on polished surface; a group of slip lines forms a slip band. Slip leaves the lattice orientation unchanged.

Twinning

* Part of the crystal shears so that the lattice becomes a MIRROR IMAGE of the parent across a TWIN PLANE. Atom movement is less than one atomic spacing and proportional to distance from the twin plane.
* Types: mechanical twins (by deformation, BCC and HCP, e.g. Fe at low T, Zn, Mg) and annealing twins (by recrystallisation in FCC like Cu, brass, Al).

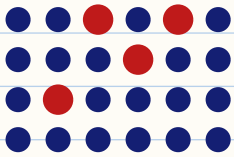
Slip	Twinning
Lattice orientation same on both sides	Orientation changes (mirror image)
Atoms move by whole lattice spacings	Movement < one atomic spacing
Takes place on widely spaced planes	Takes place on every plane in twinned region
Takes long time (ms)	Very fast (microseconds)
Needs lower stress, main mode	Needs higher stress, secondary mode

8. Constitution of Alloys

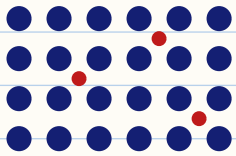
Necessity of alloying

* Pure metals are soft, weak, costly and have limited properties. ALLOY = metallic substance made of two or more elements, at least one a metal.
* Alloying is done to (i) increase strength, hardness, (ii) improve corrosion / oxidation resistance, (iii) improve casting, machining, weldability, (iv) get special magnetic / electrical / high temperature properties, (v) lower cost, (vi) allow heat treatment.
* Terms: Component = element or compound making the alloy. Phase = physically distinct, homogeneous, mechanically separable portion with its own boundary. System = group of alloys of same components. Microstructure = arrangement of phases seen under microscope.

Types of solid solutions



Substitutional



Interstitial

* SUBSTITUTIONAL solid solution: solute atoms replace solvent atoms at lattice sites (Cu-Ni, Cu-Zn). May be ordered or random.
* INTERSTITIAL solid solution: small solute atoms (H, C, N, B, O, radius < 0.59 x solvent) occupy voids between solvent atoms (C in Fe: ferrite, austenite).
* Terminal (primary) solid solution = based on one pure component; Intermediate solid solution = new structure, composition in the middle of diagram.

Hume-Rothery's rules (for extensive substitutional solid solubility)

* 1. ATOMIC SIZE FACTOR: difference in atomic radii must be less than 15%. Greater difference -> lattice strain, limited solubility.
* 2. CRYSTAL STRUCTURE: both elements should have the same crystal structure.
* 3. ELECTRO-NEGATIVITY (chemical affinity): small difference. Large difference -> compound formation instead of solution.
* 4. RELATIVE VALENCY: a metal of lower valency dissolves more easily in a metal of higher valency than vice-versa.

Cu-Ni satisfies all four rules (radius 0.128 / 0.125 nm, both FCC, similar electronegativity, valency +2) -> complete solubility.

Intermediate alloy phases

* Formed when two metals differ in chemical nature; the crystal structure is different from either component; usually hard and brittle; composition range narrow.
* (a) Interstitial compounds: transition metal + small atoms H, C, N, B (e.g. Fe3C, TiC, WC, Fe4N, TiN). Very hard. Hagg's rule: $r_x/r_M < 0.59$ gives simple structure.
* (b) Valency compounds (normal): follow chemical valency rules, e.g. Mg2Pb, Mg2Sn, Mg2Si, ZnS.
* (c) Size factor compounds: Laves phases AB2 (MgCu2, Mg2n2) - atomic radius ratio about 1.2; Sigma phase (Fe-Cr) - hard and brittle, harmful in stainless steels.

Electron compounds (Hume-Rothery phases)

* Intermediate phases whose crystal structure depends on the ratio of VALENCE ELECTRONS to ATOMS (e/a).

e/a ratio	Structure	Example
3/2 (21/14)	BCC (beta phase)	Cu2n, Cu5Sn, CuBe, Cu3Al
21/13	Complex cubic (gamma)	Cu52n8, Cu9Al4, Cu31Sn8
7/4 (21/12)	HCP (epsilon)	Cu2n3, Cu3Sn, Cu3Si

* Electron compounds are usually hard and brittle, with metallic bonding; they often have a wide composition range and show disorder-order transformation.

9. Equilibrium (Phase) Diagrams

* A phase diagram is a graph of TEMPERATURE vs COMPOSITION showing the phases present at equilibrium (very slow cooling). It is also called equilibrium / constitutional diagram.

Gibbs Phase rule

$$F = C - P + 2 \text{ (at constant pressure : } F = C - P + 1)$$

* F = degrees of freedom (variance), C = number of components, P = number of phases. For a binary alloy C = 2, so F = 3 - P. In a single phase region F = 2 (T and composition can be changed). Two-phase region F = 1. Three phase (invariant) point F = 0.

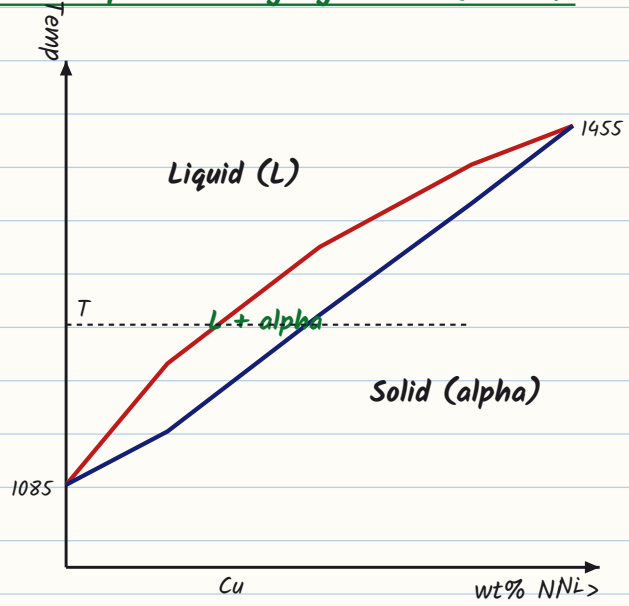
Experimental methods of construction

* 1. THERMAL ANALYSIS (cooling curves): a series of alloys is melted and cooled slowly, temperature noted with time. Change of slope / thermal arrest marks liquidus, solidus and

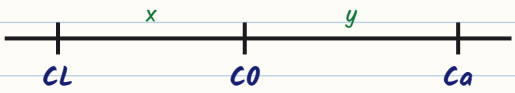
invariant temperatures. Most important method.

* 2. Microscopic examination: specimens quenched from various temperatures and the phases observed after polishing and etching.
* 3. X-ray diffraction: gives crystal structure and lattice parameter of the phases; solubility limits found from change of lattice parameter.
* 4. Dilatometry (change of length), electrical resistivity, magnetic and hardness measurements: detect solid state transformations.
* The points obtained from all alloys are plotted on T vs composition axes and joined: LIQUIDUS (start of freezing), SOLIDUS (end of freezing), solvus (solid solubility limit).

A. Isomorphous alloy systems (Cu-Ni)



Lever rule at temperature T :



W(liq) = (Ca - C0)/(Ca - CL)
W(alpha) = (C0 - CL)/(Ca - CL)
W(liq) + W(alpha) = 1

* Two metals completely soluble in BOTH liquid and solid states -> single solid phase alpha over whole range. Examples: Cu-Ni, Au-Ag, Ge-Si, Bi-Sb, Ni-Co.
* Three regions: Liquid (above liquidus), Liquid + alpha (between), alpha solid (below solidus). Melting points: Cu 1085 C, Ni 1455 C.

Equilibrium cooling of an alloy

* Alloy of composition C_0 cooled slowly: liquid cools till liquidus, first solid alpha (rich in Ni) forms. On further cooling, the composition of solid follows solidus and the liquid follows liquidus; the solid keeps changing composition by DIFFUSION so that at the solidus the entire alloy is uniform alpha of composition C_0. Heating is the reverse sequence.

Lever rule (to find amount of phases)

* Draw a horizontal TIE LINE at the temperature through the alloy composition. Its ends

give the compositions of the two phases. Amount of a phase = length of the lever ARM OPPOSITE to it / total tie-line length.

$$W_L = (C_a - C_0) / (C_a - C_L) \quad W_a = (C_0 - C_L) / (C_a - C_L)$$

Eg: 35 wt% Ni alloy at 1250 C: tie line gives C_L = 31.5% Ni and C_a = 42.5% Ni -> W_L = (42.5-35)/(42.5-31.5) = 0.68 (68% liquid) and W_a = (35-31.5)/(42.5-31.5) = 0.32 (32% solid).

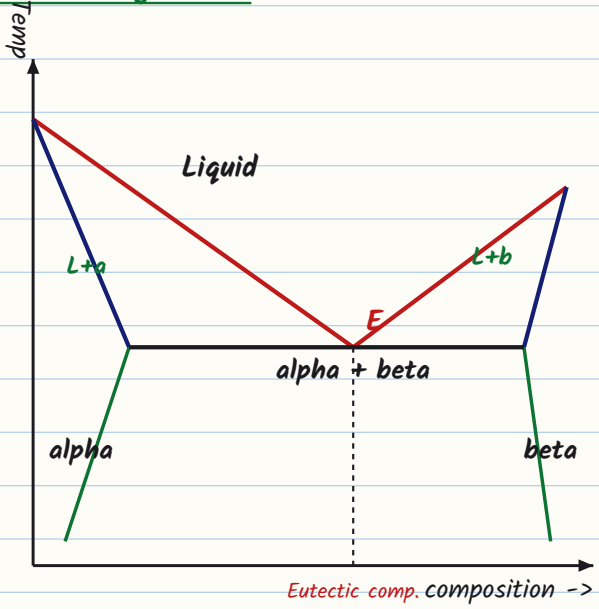
Coring (dendritic segregation)

* In NON-equilibrium (fast) cooling diffusion in the solid is too slow: centre of each grain is rich in the high melting component (Ni) and outer layers are rich in Cu. This cored structure causes variation in properties, lower melting point (hot shortness) and corrosion.
* Removed by HOMOGENIZING anneal (heating just below solidus for long time).

Miscibility gaps

* In some isomorphous systems, solid solution separates into two solid solutions of different composition at lower temperature (dome shaped region). Example: Au-Ni, Cu-Pt, Cr-W. Liquidus and solidus may show a minimum or maximum (azeotrope-like).

B. Eutectic systems



Eutectic reaction :

L (C_E) --cooling--> a + b

Invariant: F = 0 (3 phases in equilibrium, constant T)

Hypo-eutectic : C < C_E

Hyper-eutectic: C > C_E

Eg: Pb-Sn (61.9% Sn, 183 C)

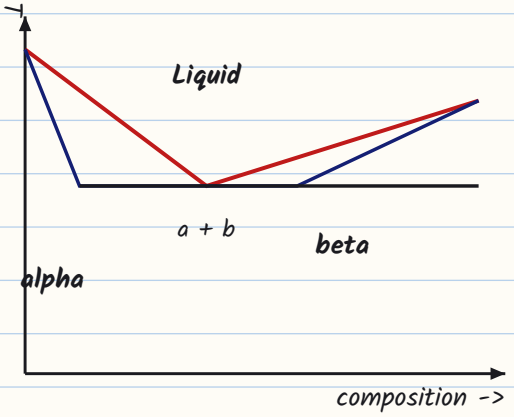
* Two components completely soluble in liquid but only PARTIALLY soluble in the solid state. Lowest melting alloy = EUTECTIC composition, which freezes at a single temperature like a pure metal.
* Eutectic alloy structure: fine mixture (lamellar) of alpha + beta. Hypoeutectic: primary alpha (proeutectic) + eutectic. Hypereutectic: primary beta + eutectic.

* Examples: Pb-Sn (solder), Al-Si, Cu-Ag, Pb-Sb (type metal), Fe-Fe3C (ledeburite).
* Solvus lines show decreasing solid solubility with fall in temperature -> basis of PRECIPITATION (age) hardening.

C. Congruent melting intermediate phases

* An intermediate compound that melts at a single temperature without change in composition (like a pure metal) is said to melt CONGRUENTLY, e.g. Mg2Pb, Mg2Sn in Mg-Pb. The diagram looks like TWO eutectic diagrams placed side by side with the compound's maximum in between.
* If the compound decomposes on heating into liquid of different composition + another solid, it melts INCONGRUENTLY (via a peritectic reaction).

D. Peritectic reaction



Peritectic reaction :

L + a (solid) ---> b (solid)

at constant T, F = 0

Eg: Fe-C at 1493 C: L + d -> g

Eutectoid : g -> a + Fe3C (727 C)

Peritectoid: a + b -> g (solid)

Monotectic: L1 -> L2 + solid

* Liquid + solid 1 ---> solid 2 on cooling at a constant temperature. Examples: Fe-C (1493 C), Cu-Zn, Pt-Ag, Fe-Ni.
* Peritectic reaction is slow; the new solid forms as a layer around the first solid, stopping diffusion -> a non-equilibrium (cored) structure usually results.

E. Transformations in the solid state

* ALLOTROPY (polymorphism): change of crystal structure in solid with temperature, e.g. Fe: BCC alpha (below 912 C) -> FCC gamma (912-1394 C) -> BCC delta (1394-1538 C). Also Ti (HCP -> BCC at 882 C), Sn (grey/white).
* EUTECTOID reaction: one solid phase -> two other solid phases on cooling. gamma (0.77% C) -> alpha + Fe3C (PEARLITE) at 727 C.
* PERITECTOID reaction: two solids -> one new solid on cooling: alpha + beta -> gamma. (Cu-Si, Ag-Al)
* Order-disorder transformation: e.g. CuZn (beta brass) above 460 C random, below it ordered.

Reaction	On cooling at constant T
Eutectic	Liquid -> solid 1 + solid 2
Eutectoid	Solid 1 -> solid 2 + solid 3
Peritectic	Liquid + solid 1 -> solid 2
Peritectoid	Solid 1 + solid 2 -> solid 3
Monotectic	Liquid 1 -> solid + liquid 2

Relationship between phase diagram and properties

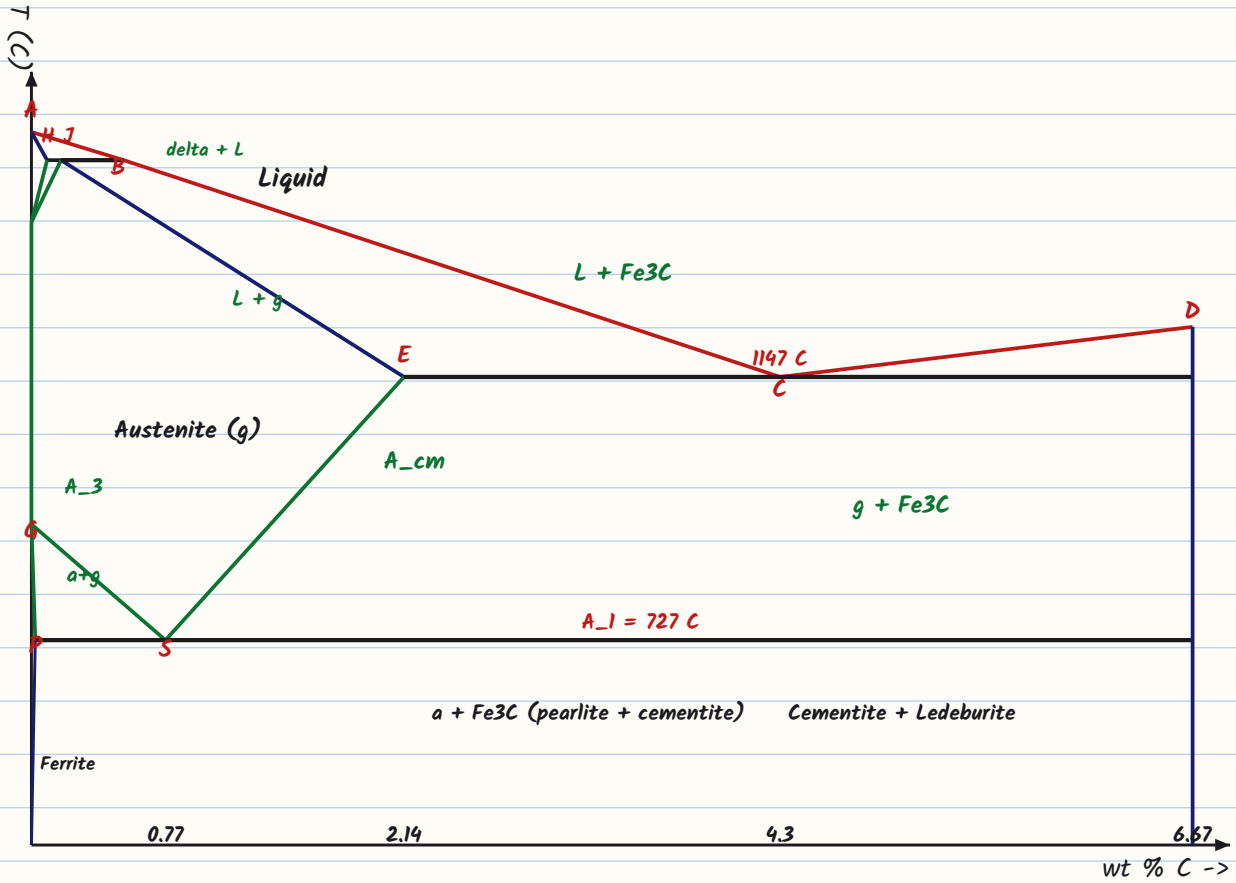
* Solid solution alloys (isomorphous): hardness / strength rise to a maximum near 50:50 composition; electrical conductivity falls to a minimum there (curved).
* Eutectic type (mechanical mixtures): properties vary LINEARLY with composition between the properties of the two phases.
* Compound formation: sharp peak (maximum hardness) at compound composition. Casting: eutectic alloys have the best fluidity and low shrinkage (good castability); wide freezing range alloys are poor in castability and prone to segregation.

10. Study of Binary Phase Diagrams

(i) Cu-Ni diagram - summary

* Isomorphous: single FCC alpha solid solution at all compositions. Liquidus and solidus lens shaped from 1085 C (Cu) to 1455 C (Ni).
* Properties: Cu-Ni alloys have excellent corrosion resistance and ductility; Monel (67 Ni - 30 Cu), Constantan (45 Ni - 55 Cu) used for resistors and thermocouples; cupro-nickel (70 Cu - 30 Ni) for marine condenser tubes and coins.
* Cannot be hardened by heat treatment, only by cold work (strain hardening) and solid solution strengthening.

(ii) Iron - Iron carbide (Fe-Fe3C) diagram



* Iron-carbon alloys up to 6.67% C (the compound Fe3C, CEMENTITE). Fe3C is metastable but treated as an equilibrium phase. Alloys < 2.14% C are STEELS; 2.14% - 6.67% C are CAST IRONS.

Phases

* FERRITE (alpha): interstitial solid solution of C in BCC iron; max 0.022% C at 727 C; soft, ductile, magnetic below 768 C (Curie temp.). delta-ferrite: BCC at high temperature (up to 0.09% C at 1493 C).
* AUSTENITE (gamma): interstitial solid solution of C in FCC iron; max 2.14% C at 1147 C; soft, ductile, non-magnetic; stable only above 727 C.
* CEMENTITE (Fe3C): intermetallic compound with 6.67% C, orthorhombic; very hard and brittle.
* PEARLITE: eutectoid mixture of alternate lamellae of ferrite + cementite (0.77% C), strong with fair ductility. LEDEBURITE: eutectic mixture of austenite + cementite (4.3% C, 1147 C); transformed ledeburite (pearlite + cementite) below 727 C.

Invariant reactions

Reaction	Temp.	Comp. (% C)	Equation
Peritectic	1493 C	0.17	L + delta-ferrite -> austenite
Eutectic	1147 C	4.3	L -> austenite + Fe3C
Eutectoid	727 C	0.77	austenite -> ferrite + Fe3C

Critical temperature lines

* A1 (727 C): eutectoid line, PSK. A3: GS line - ferrite begins to form from austenite (912 C for pure iron). Acm: SE line - cementite begins to separate from austenite in hypereutectoid steels. A2 : 768 C (magnetic change).

Microstructures on slow cooling

* EUTECTOID steel (0.77% C): all austenite -> 100% pearlite.
* HYPOEUTECTOID steel (< 0.77% C): proeutectoid FERRITE (white) + pearlite.
* HYPEREUTECTOID steel (0.77 - 2.14% C): proeutectoid CEMENTITE network along grain boundaries + pearlite.
* Cast irons: hypoeutectic (2.14-4.3% C): pearlite + cementite + transformed ledeburite; eutectic (4.3%): ledeburite; hypereutectic (4.3-6.67%): primary cementite + ledeburite.

Worked example (lever rule at just below 727 C)

Hypoeutectoid steel with 0.4% C:

% proeutectoid ferrite = (0.77 - 0.4)/(0.77 - 0.022) x 100 = 49.5%

% pearlite = (0.4 - 0.022)/(0.77 - 0.022) x 100 = 50.5%

Total ferrite (in ferrite + pearlite) = (6.67 - 0.4)/(6.67 - 0.022) = 94.3%, cementite = 5.7%.

Hypereutectoid 1.2% C steel: proeutectoid cementite = (1.2 - 0.77)/(6.67 - 0.77) = 7.3%; pearlite = 92.7%.

Quick revision

* APF: SC 0.52, BCC 0.68, FCC = HCP = 0.74. Slip systems: FCC 12, BCC 12+, HCP 3.
* Hume-Rothery rules: size < 15%, same structure, similar electronegativity, higher valency solvent.
* Phase rule F = C - P + 1. Invariant reactions: eutectic, peritectic, eutectoid, peritectoid, monotectic.
* Fe-Fe3C: 727 C eutectoid (0.77% C), 1147 C eutectic (4.3% C), 1493 C peritectic (0.17% C); steels < 2.14% C.